

GUDIPATI SADASHIV SHRI RAM

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SUMMARY:

Mechanical Design Engineer experienced in CAD, prototyping, and fabrication, with expertise in designing and testing mechanical systems, fixtures, enclosures, and structural components.

TECHNICAL SKILLS:

Mechanical Design & Product Development: Product Design, Mechanical System Design, DFM/DFA, GD&T (ASME Y14.5), Tolerance Analysis, Tolerance Stack-Up Analysis, Engineering Drawings, Manufacturing Drawings, BOM Management, Design Verification & Validation (DV&V), Prototype Development.

CAD & Engineering Tools: SolidWorks (CSWA & CSWP), Siemens NX, Fusion 360, AutoCAD, Parametric Modeling, Assembly Design, Sheet Metal Design, Weldment Design.

Analysis & Simulation: ANSYS Workbench, COMSOL Multiphysics, FEA, Structural Analysis, Thermal Analysis, CFD, Design Validation.

Manufacturing & Fabrication: CNC Machining, Additive Manufacturing (FDM, SLA/Resin), Sheet Metal Fabrication, Prototype Testing, TIG/MIG Welding, Manufacturing Processes.

Electromechanical Systems & Technical Tools: Raspberry Pi, Arduino, PCB Design, Sensors & Actuators, MATLAB, Python, C/C++, Git.

TECHNICAL EXPERIENCE:

MACLAB – Matter Assembly Computation Lab | CU Boulder

Aug 2025 – May 2026

- Designed compliant mechanical structures for capacitive sensing applications, optimizing deformation response under load
- Performed mechanical modeling and experimental validation of stress-strain behaviour in multi-material systems
- Developed a precision rotary encoder ($\pm 1^\circ$ resolution) with emphasis on tolerance, repeatability, and compact design.

Assistant Engineering Supervisor |GVSM Enterprises, Vijayawada

Jan 2022 – May 2023

- Designed and developed custom mechanical fixtures for handling specialized loads, incorporating load calculations, material selection, and safety factors to ensure structural integrity and operational efficiency.
- Supervised fabrication and assembly of fixtures, coordinating with shop-floor technicians, validating dimensional tolerances, and ensuring adherence to engineering drawings and manufacturing standards.
- Conducted preventive and corrective maintenance planning for diesel generator systems and fleet vehicles, performing fault diagnostics and coordinating repairs to maintain high equipment uptime.
- Performed material quantity take-offs and developed cost estimation models for fixture design and fabrication, including raw material selection, machining processes, and vendor coordination, ensuring alignment with project timelines and budget constraints.

In-Plant Engineering Intern |Ordnance Factory Medak (Ministry of Defense, Gov of India)

Jul 2021 – Dec 2021

- Designed and modeled an automated mechanical door actuation mechanism for armored vehicle applications using Siemens NX, focusing on linkage geometry and range-of-motion constraints
- Analyzed manufacturing and assembly processes for BMP Sarath infantry combat vehicles, gaining exposure to precision machining, tolerance stack-ups, and large-scale mechanical assembly workflows
- Documented component assembly sequences and manufacturing process flows for defense-grade mechanical systems.

EDUCATION:

Master of Science, Mechanical Engineering (Robotics & Controls) |University of Colorado, Boulder, CO, USA

Jan 2024 – May 2026

Coursework: Advanced Dynamics, Feedback Control, Linear Systems, Introduction to Robotics, Micro Electromechanical Systems, Robust Multivariable Controls, Systems Engineering, Independent Study (Tactile sensors).

Bachelor of Engineering in Mechanical Engineering |Osmania University, Hyderabad, Telangana, India

Aug 2017 – July 2021

PROJECTS:

Transient-Wing Tilt-Rotor Drone

- Designed and developed a tilt-rotor VTOL UAV integrating 4x BLDC motors, ESCs, Li-ion battery systems, and embedded flight control.
- Engineered tilt-wing actuation mechanism with servo-driven transition between hover and forward flight modes.
- Performed structural design and weight optimization using additive manufacturing and CAD-based analysis.
- Integrated IMU, GPS, and RF communication modules for real-time flight control and telemetry.
- Implemented hardware-software integration for control loops and flight stability.

Custom UAV Ground Control Station (Raspberry Pi)

- Designed and built a portable Raspberry Pi-based UAV ground control station with dual joystick controls, NRF24L01 RF link, NEO-6M GPS, and 5.8 GHz FPV receiver for real-time flight monitoring.
- Designed and fabricated a handheld enclosure and mechanical layout for a portable ground control system
- Integrated joystick mechanisms, display housing, and internal component packaging for ergonomic and structural stability
- Developed a modular mechanical design to support rapid iteration and component accessibility

LiDAR-Based Obstacle Avoidance System

- Designed mechanical integration and mounting architecture for LiDAR sensors and actuation components
- Developed a compact system layout ensuring sensor stability and alignment during operation
- Supported integration of sensing and actuation systems within a mobile robotic platform

Quadrotor Dynamics with Wind Gusts

- Simulated PID and MPC controllers in MATLAB/Simulink, improving stability by 40% under wind disturbances.
- Applied Extended Kalman Filter (EKF) for state estimation and sensor fusion, reducing trajectory error.
- Explored nonlinear control for aggressive maneuvers and disturbance rejection.

Automatic Vehicle Detection for Self-Driving Cars – IJARESM, Vol. 11, Issue 8, Aug 2023, ISSN: 2455-6211

- Applied deep learning models (YOLO, Faster-RCNN, SSD) using TensorFlow to improve autonomous vehicle perception, focusing on detection of pedestrians, vehicles, and traffic signals to enhance road safety.